

ADM-HQ:

- Устанавливаем Terraform:
 - из-под суперпользователя **root**

```
wget https://hashicorp-releases.yandexcloud.net/terraform/1.14.5/terraform\_1.14.5\_linux\_amd64.zip
```

```
unzip terraform_1.14.5_linux_amd64.zip -d /usr/local/bin/
```

- Проверить:
 - из-под системного пользователя **user**

```
[user@adm-hq ~]$ terraform --version
Terraform v1.14.5
on linux_amd64
```

- Настраиваем Terraform:
 - из-под системного пользователя **user**

```
cat <<EOF > ~/.terraformrc
provider_installation {
  network_mirror {
    url = "https://terraform-mirror.mcs.mail.ru"
    include = ["registry.terraform.io/*/*"]
  }
  direct {
    exclude = ["registry.terraform.io/*/*"]
  }
}
EOF
```

- Создаём требуемую директорию для хранения файлов Terraform:

```
mkdir /home/user/terraform
cd /home/user/terraform
```

- Создаём файл **terraform.tf** и указываем каких провайдеров будем использовать:

```
cat <<EOF > terraform.tf
terraform {
  required_providers {
    freeipa = {
      source = "camptocamp/freeipa"
      version = "1.0.0"
    }
  }
}
EOF
```

- Создаём файл **providers.tf** и указываем необходимые для подключения параметры для каждого из провайдеров:

```
cat <<EOF > providers.tf
provider "freeipa" {
  host      = var.freeipa_host
  username  = var.freeipa_username
  password  = var.freeipa_username_password
  insecure  = true
}
EOF
```

- Создаём файл **variable.tf** и указываем переменные которые будет использовать Terraform:

```
cat <<EOF > variable.tf
variable "freeipa_host" {
  type      = string
  description = "Access to the FreeIPA host"
}

variable "freeipa_username" {
  type      = string
  description = "Access to the FreeIPA host username"
}

variable "freeipa_username_password" {
  type      = string
  description = "Access to the FreeIPA host username password"
  sensitive  = true
}
EOF
```

- Создаём файл **terraform.tfvars** и указываем значения переменных которые были указаны в файле **variable.tf**:

```
cat <<EOF > terraform.tfvars
freeipa_host      = "srv-hq.au.team"
freeipa_username  = "admin"
freeipa_username_password = "P@ssw0rd"
EOF
```

- Инициализируем директорию для работы с Terraform и указанным провайдером:

```
terraform init
```

- должно получиться следующее:

```
...
Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

- Проверить:

```
[user@adm-hq terraform]$ tree -a
.
├── providers.tf
├── .terraform
│   └── providers
│       └── registry.terraform.io
│           └── camptocamp
│               └── freeipa
│                   └── 1.0.0
│                       └── linux_amd64
│                           ├── CHANGELOG.md
│                           ├── LICENSE
│                           ├── README.md
│                           └── terraform-provider-freeipa_v1.0.0
├── .terraform.lock.hcl
├── terraform.tf
├── terraform.tfvars
└── variable.tf

8 directories, 9 files
[user@adm-hq terraform]$
```

- Реализуем необходимый функционал:

```
cat <<EOF >> variable.tf
```

```
variable "reverse_zones" {
  description = "List of reverse viewing zones"
  type        = list(string)
  default = [
    "2.1.10.in-addr.arpa.",
    "0.2.10.in-addr.arpa.",
    "1.2.10.in-addr.arpa.",
    "2.2.10.in-addr.arpa.",
    "16.172.in-addr.arpa."
  ]
}

variable "dns_records" {
  description = "List of DNS records (A and corresponding PTR)"
  type = list(object({
    hostname          = string
    ip_address        = string
    forward_zone      = string
    reverse_zone       = optional(string)
    reverse_zone_record = optional(string)
  }))
  default = [
    {
      hostname          = "fw-hq"
      ip_address        = "10.1.1.1"
      forward_zone      = "au.team."
      reverse_zone       = "1.1.10.in-addr.arpa."
      reverse_zone_record = "1"
    },
    {
      hostname          = "adm-hq"
      ip_address        = "10.1.1.46"
      forward_zone      = "au.team."
      reverse_zone       = "1.1.10.in-addr.arpa."
      reverse_zone_record = "46"
    },
    {
      hostname          = "rtr-br"
      ip_address        = "10.2.0.1"
      forward_zone      = "au.team."
      reverse_zone       = "0.2.10.in-addr.arpa."
      reverse_zone_record = "1"
    },
    {
      hostname          = "fw-br"
      ip_address        = "10.2.0.2"
      forward_zone      = "au.team."
      reverse_zone       = "0.2.10.in-addr.arpa."
      reverse_zone_record = "2"
    },
    {
      hostname          = "srv-br"
      ip_address        = "10.2.1.10"
      forward_zone      = "au.team."
      reverse_zone       = "1.2.10.in-addr.arpa."
      reverse_zone_record = "10"
    }
  ]
}
```

```

},
{
    hostname          = "rtr-cod"
    ip_address         = "172.16.1.254"
    forward_zone       = "au.team."
    reverse_zone       = "16.172.in-addr.arpa."
    reverse_zone_record = "254.1"
},
{
    hostname          = "sw-cod"
    ip_address         = "172.16.1.0"
    forward_zone       = "au.team."
    reverse_zone       = "16.172.in-addr.arpa."
    reverse_zone_record = "0.1"
},
{
    hostname          = "ha1-cod"
    ip_address         = "172.16.0.1"
    forward_zone       = "au.team."
    reverse_zone       = "16.172.in-addr.arpa."
    reverse_zone_record = "1.0"
},
{
    hostname          = "ha2-cod"
    ip_address         = "172.16.0.2"
    forward_zone       = "au.team."
    reverse_zone       = "16.172.in-addr.arpa."
    reverse_zone_record = "2.0"
},
{
    hostname          = "srv1-cod"
    ip_address         = "172.16.1.1"
    forward_zone       = "au.team."
    reverse_zone       = "16.172.in-addr.arpa."
    reverse_zone_record = "1.1"
},
{
    hostname          = "srv2-cod"
    ip_address         = "172.16.1.2"
    forward_zone       = "au.team."
    reverse_zone       = "16.172.in-addr.arpa."
    reverse_zone_record = "2.1"
},
{
    hostname          = "srv3-cod"
    ip_address         = "172.16.1.3"
    forward_zone       = "au.team."
    reverse_zone       = "16.172.in-addr.arpa."
    reverse_zone_record = "3.1"
}
]
}
EOF

```

- создаём файл **dns.tf** и указываем требуемые ресурсы

```

resource "freeipa_dns_zone" "reverse" {
  for_each = toset(var.reverse_zones)
  zone_name = each.value
}

resource "freeipa_dns_record" "a" {
  for_each = { for r in var.dns_records : r.hostname => r }
  dnszoneidnsname = each.value.forward_zone
  idnsname = each.value.hostname
  records = [each.value.ip_address]
  type = "A"
}

resource "freeipa_dns_record" "ptr" {
  for_each = { for r in var.dns_records : r.hostname => r }
  dnszoneidnsname = each.value.reverse_zone
  idnsname = each.value.reverse_zone_record
  records = ["${each.value.hostname}.${each.value.forward_zone}"]
  type = "PTR"

  depends_on = [freeipa_dns_zone.reverse]
}

```

- Запускаем развёртывание ресурсов:

```
terraform apply -auto-approve
```

- должно получиться следующее:

```

Apply complete! Resources: 29 added, 0 changed, 0 destroyed.
[user@adm-hq terraform]$ █

```

- Проверить:

```
[user@adm-hq terraform]$ terraform state list
freeipa_dns_record.a["adm-hq"]
freeipa_dns_record.a["fw-br"]
freeipa_dns_record.a["fw-hq"]
freeipa_dns_record.a["ha1-cod"]
freeipa_dns_record.a["ha2-cod"]
freeipa_dns_record.a["rtr-br"]
freeipa_dns_record.a["rtr-cod"]
freeipa_dns_record.a["srv-br"]
freeipa_dns_record.a["srv1-cod"]
freeipa_dns_record.a["srv2-cod"]
freeipa_dns_record.a["srv3-cod"]
freeipa_dns_record.a["sw-cod"]
freeipa_dns_record.ptr["adm-hq"]
freeipa_dns_record.ptr["fw-br"]
freeipa_dns_record.ptr["fw-hq"]
freeipa_dns_record.ptr["ha1-cod"]
freeipa_dns_record.ptr["ha2-cod"]
freeipa_dns_record.ptr["rtr-br"]
freeipa_dns_record.ptr["rtr-cod"]
freeipa_dns_record.ptr["srv-br"]
freeipa_dns_record.ptr["srv1-cod"]
freeipa_dns_record.ptr["srv2-cod"]
freeipa_dns_record.ptr["srv3-cod"]
freeipa_dns_record.ptr["sw-cod"]
freeipa_dns_zone.reverse["0.2.10.in-addr.arpa."]
freeipa_dns_zone.reverse["1.2.10.in-addr.arpa."]
freeipa_dns_zone.reverse["16.172.in-addr.arpa."]
freeipa_dns_zone.reverse["2.1.10.in-addr.arpa."]
freeipa_dns_zone.reverse["2.2.10.in-addr.arpa."]
[user@adm-hq terraform]$
```

- При повторном запуске, ошибок не возникает:

```
[user@adm-hq terraform]$ terraform apply -auto-approve
freeipa_dns_record.a["sw-cod"]: Refreshing state...
freeipa_dns_zone.reverse["16.172.in-addr.arpa."]: Refreshing state... [id=16.172.in-addr.arpa.]
freeipa_dns_zone.reverse["2.1.10.in-addr.arpa."]: Refreshing state... [id=2.1.10.in-addr.arpa.]
freeipa_dns_record.a["adm-hq"]: Refreshing state...
freeipa_dns_record.a["ha1-cod"]: Refreshing state...
freeipa_dns_record.a["srv1-cod"]: Refreshing state...
freeipa_dns_zone.reverse["1.2.10.in-addr.arpa."]: Refreshing state... [id=1.2.10.in-addr.arpa.]
freeipa_dns_record.a["rtr-br"]: Refreshing state...
freeipa_dns_zone.reverse["2.2.10.in-addr.arpa."]: Refreshing state... [id=2.2.10.in-addr.arpa.]
freeipa_dns_zone.reverse["0.2.10.in-addr.arpa."]: Refreshing state... [id=0.2.10.in-addr.arpa.]
freeipa_dns_record.a["ha2-cod"]: Refreshing state...
freeipa_dns_record.a["fw-hq"]: Refreshing state...
freeipa_dns_record.a["srv-br"]: Refreshing state...
freeipa_dns_record.a["srv2-cod"]: Refreshing state...
freeipa_dns_record.a["srv3-cod"]: Refreshing state...
freeipa_dns_record.a["fw-br"]: Refreshing state...
freeipa_dns_record.a["rtr-cod"]: Refreshing state...
freeipa_dns_record.ptr["srv1-cod"]: Refreshing state...
freeipa_dns_record.ptr["ha1-cod"]: Refreshing state...
freeipa_dns_record.ptr["sw-cod"]: Refreshing state...
freeipa_dns_record.ptr["ha2-cod"]: Refreshing state...
freeipa_dns_record.ptr["srv-br"]: Refreshing state...
freeipa_dns_record.ptr["adm-hq"]: Refreshing state...
freeipa_dns_record.ptr["rtr-br"]: Refreshing state...
freeipa_dns_record.ptr["srv2-cod"]: Refreshing state...
freeipa_dns_record.ptr["rtr-cod"]: Refreshing state...
freeipa_dns_record.ptr["srv3-cod"]: Refreshing state...
freeipa_dns_record.ptr["fw-hq"]: Refreshing state...
freeipa_dns_record.ptr["fw-br"]: Refreshing state...
```

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

```
[user@adm-hq terraform]$
```

- Проверить работоспособность:

```
[user@adm-hq terraform]$ host fw-hq.au.team
fw-hq.au.team has address 10.1.1.1
[user@adm-hq terraform]$ host adm-hq.au.team
adm-hq.au.team has address 10.1.1.46
[user@adm-hq terraform]$ host rtr-br.au.team
rtr-br.au.team has address 10.2.0.1
[user@adm-hq terraform]$ host fw-br.au.team
fw-br.au.team has address 10.2.0.2
[user@adm-hq terraform]$ host srv-br.au.team
srv-br.au.team has address 10.2.1.10
[user@adm-hq terraform]$ host rtr-cod.au.team
rtr-cod.au.team has address 172.16.1.254
[user@adm-hq terraform]$ host sw-cod.au.team
sw-cod.au.team has address 172.16.1.0
[user@adm-hq terraform]$ host ha1-cod.au.team
ha1-cod.au.team has address 172.16.0.1
[user@adm-hq terraform]$ host ha2-cod.au.team
ha2-cod.au.team has address 172.16.0.2
[user@adm-hq terraform]$ host srv1-cod.au.team
srv1-cod.au.team has address 172.16.1.1
[user@adm-hq terraform]$ host srv2-cod.au.team
srv2-cod.au.team has address 172.16.1.2
[user@adm-hq terraform]$ host srv3-cod.au.team
srv3-cod.au.team has address 172.16.1.3
[user@adm-hq terraform]$ host 10.1.1.1
1.1.1.10.in-addr.arpa domain name pointer fw-hq.au.team.
[user@adm-hq terraform]$ host 10.1.1.46
46.1.1.10.in-addr.arpa domain name pointer adm-hq.au.team.
[user@adm-hq terraform]$ host 10.2.0.1
1.0.2.10.in-addr.arpa domain name pointer rtr-br.au.team.
[user@adm-hq terraform]$ host 10.2.0.2
2.0.2.10.in-addr.arpa domain name pointer fw-br.au.team.
[user@adm-hq terraform]$ host 10.2.1.10
10.1.2.10.in-addr.arpa domain name pointer srv-br.au.team.
[user@adm-hq terraform]$ host 172.16.1.254
254.1.16.172.in-addr.arpa domain name pointer rtr-cod.au.team.
[user@adm-hq terraform]$ host 172.16.1.0
0.1.16.172.in-addr.arpa domain name pointer sw-cod.au.team.
[user@adm-hq terraform]$ host 172.16.0.1
1.0.16.172.in-addr.arpa domain name pointer ha1-cod.au.team.
[user@adm-hq terraform]$ host 172.16.0.2
2.0.16.172.in-addr.arpa domain name pointer ha2-cod.au.team.
[user@adm-hq terraform]$ host 172.16.1.1
1.1.16.172.in-addr.arpa domain name pointer srv1-cod.au.team.
[user@adm-hq terraform]$ host 172.16.1.2
2.1.16.172.in-addr.arpa domain name pointer srv2-cod.au.team.
[user@adm-hq terraform]$ host 172.16.1.3
3.1.16.172.in-addr.arpa domain name pointer srv3-cod.au.team.
```